

Expected Value: The Mathematics of Betting

Game State

Community Cards



Player Hands

Your Hand:



Opponent:



Additional Content

Expected Value: The Mathematics of Betting

What is Expected Value?

Expected Value (EV) = The average outcome if you repeat an action many, many times.

Formula

$$EV = (\text{Probability of Win} \times \text{Value if Win}) + (\text{Probability of Loss} \times \text{Value if Loss})$$

Example 1: Simple Bet

****Bet****: \$10

****Win****: 20% chance to win \$50

****Lose****: 80% chance to lose \$10

****Calculation****:

- $EV = (0.20 \times \$50) + (0.80 \times -\$10)$

- $EV = \$10 + (-\$8)$

- $EV = \$2$

****Meaning****: Over many bets, you'll average a \$2 profit per bet.

Example 2: Poker Tournament

****Situation****: Final table, 3 players left

- 1st place: \$10,000 (33% chance)

- 2nd place: \$5,000 (33% chance)

- 3rd place: \$2,500 (33% chance)

****Your Expected Value**:**

- $EV = (0.33 \times \$10,000) + (0.33 \times \$5,000) + (0.33 \times \$2,500)$

- $EV = \$3,333 + \$1,667 + \$833$

- $EV = \$5,833$

****Meaning**:** Your "average" finish is worth \$5,833, even though you'll only get one of the three prizes.

Example 3: The House Edge

****Game**:** Roulette (American)

- Bet \$10 on red

- Win: 18/38 chance to win \$10 (get \$20 back)

- Lose: 20/38 chance to lose \$10

****Your EV**:**

- $EV = (18/38 \times \$10) + (20/38 \times -\$10)$

- $EV = \$4.74 + (-\$5.26)$

- $EV = -\$0.52$

****Meaning**:** Every \$10 bet costs you \$0.52 on average. That's the house edge.

Key Insight

Expected value tells you what will happen *on average* over many repetitions. But any single bet could win big or lose everything. That's the difference between probability (long-term) and luck (short-term).

The River King's Wisdom

> "The house edge isn't evil—it's mathematics. The cards don't care about your feelings. They follow the laws of probability. But you still have to play your hand."
\"